

COMPRESSED GAS CYLINDERS

Control tip-over, valve-shear, fire, explosion, leakage and wrong-gas connection hazards during cylinder receipt, storage, transport and use.

PURPOSE AND SCOPE

Control tip-over, valve-shear, fire, explosion, leakage and wrong-gas connection hazards during cylinder receipt, storage, transport and use.

KEY HAZARDS

1  VALVE SHEAR	2  TIP-OVER	3  LEAK	4  FIRE
5  OXYGEN ENRICHMENT	6  WRONG REGULATOR	7  OIL / GREASE	8  HEAT

CRITICAL VALUES TO VERIFY IN THE FIELD

6.1 m Stored oxygen-fuel gas separation	1.5 m / 30 min Alternative noncombustible barrier height / rating	100% Upright and secure restraint
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REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

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STANDARDS AND FIELD REQUIREMENTS

OSHA 1926.350 addresses upright restraint, valve caps, safe movement, oxygen-fuel gas separation and correct equipment.

OSHA 1926.350(a)

MOVEMENT AND RESTRAINT

Secure cylinders upright; use a proper cart and never lift with magnets or improvised slings.

OSHA 1926.350(a)(6)

VALVE-PROTECTION CAP

Keep caps in place and secure when regulators are not connected; never lift by the cap.

OSHA 1926.350(a)(10)

STORAGE SEPARATION

Separate oxygen from fuel gas by 6.1 m or use a noncombustible barrier at least 1.5 m high with a 30-minute fire rating.

OSHA 1926.350(h)

CORRECT EQUIPMENT

Use gas-compatible regulators, hoses and fittings; keep oxygen equipment free of oil and grease.

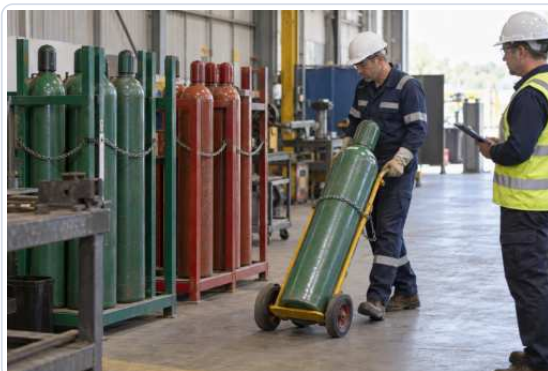
MINIMUM EVIDENCE BEFORE START

- | | |
|--|---|
| ☐ Cylinder inventory / identity check | ☐ Storage-area layout |
| ☐ Cylinder and regulator inspection | ☐ Cylinder-cart inspection |
| ☐ Leak-test method | ☐ User training record |

SUPERVISOR DISCUSSION POINTS

- Which gas is in the cylinder?
- How are oxygen and fuel gas separated?
- What steps are required before transport?

UPRIGHT RESTRAINT, SEGREGATION AND CART TRANSPORT



Identity - Cap - Restraint - Separation - Correct cart - Correct regulator

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CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. REMOVE UNNEEDED CYLINDERS

Keep only cylinders required for the task and control full/empty status.

2. SEPARATE AND SECURE

Secure upright, segregate by compatibility and protect from heat and damage.

3. MOVE WITH A CYLINDER CART

Close the valve, remove the regulator, fit the cap and move on a chained cylinder cart.

4. CORRECT REGULATOR AND LEAK CONTROL

Use the correct regulator and flashback protection; leak-test using an approved method.

CRITICAL WARNING



A sheared cylinder valve can turn the cylinder into an uncontrolled projectile. Never roll, drag or lift a cylinder by its cap.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|---|--|
| ☐ Cylinder identity readable | ☐ Inspection date current |
| ☐ Upright and secure | ☐ Cap installed |
| ☐ Gases correctly separated | ☐ Regulator compatible |
| ☐ Hoses intact | ☐ No leakage |
| ☐ Cart and chain ready | |

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ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			